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Perspectives from a Workshop: Intelligent Assessment in the age of Artificial Intelligence

Abstract

The advent of Generative Artificial Intelligence (GenAI) is already impacting pedagogic strategies and assessment methodologies in higher education, particularly in the biological sciences which have traditionally relied heavily on written assessments. GenAI's rapid and plausible text generation capabilities challenge traditional written assessments and prompt a shift towards more authentic assessment types. This paper explores innovative applications of GenAI in biology education through case studies presented at a recent workshop. These case studies illustrate how GenAI has the potential to enhance academic activities, from developing learning resources to fostering student engagement through active learning strategies. The discussion highlights a shift from product-oriented assessments to process-oriented approaches that prioritize continuous interaction, iteration, and reflection among learners. Despite GenAI's reliance on pre-existing data raising concerns about originality and contextual accuracy, and its limitations in tasks requiring high creativity and deep understanding, it has the potential to enhance educational practices when applied with awareness of its constraints. The paper concludes with a balanced analysis of the transformative impact and inherent challenges of integrating GenAI into biology education, advocating for thoughtful implementation to ensure it augments rather than replaces traditional teaching methods.

New and Noteworthy

Generative AI is transforming higher education by enabling rapid learning resource development, enhancing student engagement, and supporting authentic assessment. Our workshop-based case studies, highlight GenAI's ability to foster interactive, process-oriented learning in biosciences while addressing challenges with creativity and originality. From creating tailored quizzes to promoting active learning and ethical AI use, these strategies empower educators to integrate AI responsibly, ensuring it enriches teaching and learning in bioscience education while maintaining academic integrity.

Introduction

The advent of Generative Artificial Intelligence (GenAI) is driving a transformative shift in pedagogic strategies and assessment methodologies in Higher Education. GenAI is a subset of artificial intelligence, which focuses on the generation of new content, offering unprecedented capabilities in creating text, images, and even code, based on extensive training datasets (1). As educational institutions seek to leverage these advancements, the integration of Artificial Intelligence in education has emerged as a key area for pedagogic consideration (2).

A key feature of current GenAI tools is the rapid and facile generation of plausible, high quality written text. This is an issue in biology education as teaching in the biological sciences, perhaps more than in other STEM subjects, has historically relied heavily on written assessment. At the same time that we are considering redesigning assessments in the face of GenAI, there is also a realisation that we, as educators, need to provide more authentic assessment types to our students (3).

This paper presents a series of case studies from recent presentations at a workshop on GenAI in higher education. These case studies showcase innovative uses of GenAI across academic activities, including the rapid creation of learning materials and the enhancement of student engagement through active learning strategies. Rather than simply replacing traditional teaching tools, these applications point to a broader pedagogical shift: from product-oriented assessments, where success

is measured by a final output, to process-oriented learning, which values the journey of learning itself (4). This shift is particularly relevant for physiology educators, where developing critical thinking, applying knowledge in complex contexts, and refining understanding through experimentation and feedback are essential. A process-oriented approach supported by GenAI encourages students to engage in iterative learning, reflection, and continuous improvement, skills that mirror the scientific and diagnostic reasoning central to physiology and biosciences education (5). Although the examples span the biosciences, many reflect challenges and opportunities directly relevant to physiology education, such as teaching complex processes, encouraging iterative reasoning, and fostering authentic skills in scientific communication.

Despite the promising applications of GenAI, there are significant limitations that must be acknowledged. The technology's current reliance on pre-existing data for content generation raises concerns about the originality and contextual accuracy of its outputs. Furthermore, GenAI often struggles with tasks requiring high levels of creativity and deep understanding, such as the production of graphical abstracts or the interpretation of complex academic texts (6). These limitations highlight the challenges of integrating GenAI into assessments that require critical thinking and a deep understanding of subject matter. However, when applied with an awareness of these limitations, GenAI can still play a crucial role in enhancing educational practices (7).

This paper aims to provide a critical analysis of both the potential and the challenges of such technological integration. By examining the transformative impact alongside the inherent limitations of GenAI, we present to a balanced discussion on the future role of artificial intelligence in education.

Activity	Author	Summary	Impact
Introducing GenAI for Student Engagement and Literacy			
Introducing Gemini to Bioscience Students	LA	Use GenAI to scaffold brainstorming and build AI literacy through structured, critical group activities.	Integration of Gemini led to increased student confidence, faster project progress, and improved understanding of the LLM's strengths and limitations. The interactive workshop also sparked lively discussion and critical thinking.
Boosting student confidence through assessment literacy to reduce GenAI misuse	JC	Reduce GenAI misuse by building transparency, confidence, and assessment literacy.	Boosting students' confidence through assessment literacy helps reduce GenAI misuse and improve performance. Positive student feedback led to the adoption of similar sessions in other modules.
AI-generated MCQs and learning resources	PB	Save time by generating formative quizzes and rubrics while	Writing learning and teaching resources, as well as developing assessment rubrics, are

		embedding AI literacy.	time-demanding activities, where the use of GenAI can offer time-savings and help extend the scope of teaching materials.
Using GenAI to Enhance Teaching Resources and Learning Design			
Rapid prototyping of learning resources	CH	Quickly generate adaptable teaching resources to support responsive, student-led learning.	GenAI reduced workload and enabled faster, high-quality resource creation. It improved student preparedness and engagement, supporting deeper discussions and adaptive, student-led learning.
Generation of ideas for active learning	CH	Co-design engaging, problem-based activities that promote discussion and critical thinking.	Co-creating active learning materials with GenAI accelerated the design process and enabled a more creative lecture format.
Prompt engineering for lecture transcripts	CH	Convert lecture transcripts into concise, accessible summaries with prompts for self-study.	Lecture transcripts offer a quick and searchable repository of teaching content for students to find key topics. Transcripts can then be used in subsequent prompts to generate further reading or formative assessments.
Using Integrated Knowledge Prompting to develop written content	MH	Combine student knowledge with GenAI scaffolding to support inclusive and accurate written communication.	Helped reduce barriers for students with writing challenges, supported non-native speakers, and encouraged critical engagement with content. Promoted inclusivity and positioned GenAI as a scaffold rather than a substitute for learning.
Designing Authentic and AI-Resilient Assessment and Feedback			
Creation and Design of Graphical Abstracts as an Assessment	APS	Require creativity and synthesis beyond GenAI's current capabilities.	This assessment is efficient, sustainable, and adaptable, grading takes only minutes, and new tasks can be easily

			created by selecting different research papers. Crucially, it is also highly AI-resilient, as it requires students to demonstrate deep understanding and apply creative and visual design skills that current generative AI tools struggle to replicate reliably.
Comparing two approaches to SAQs: adapt or adopt to AI	DH	Balance academic integrity, student engagement, and AI resilience through adapted formats.	GenAI tools can enhance engagement and learning, but open-book formats may mask gaps in understanding. Authenticity of student work remains a concern, underscoring the need for well-aligned assessments, strong preparation, and ongoing dialogue about ethical AI use.
Using digital quizzes as formative assessments to engage and prepare students for lab reports	HC	Build student skills and confidence with low-stakes, feedback-rich tasks.	The quiz reduced marking workload, supported authentic skill development, and helped deter misuse of GenAI by building student confidence through low-stakes, feedback-driven practice.
A Lab Report Plan as a formative assessment task for a Summative Lab Report	HC	Scaffold lab report writing through staged planning and targeted feedback.	The activity deterred GenAI misuse by fostering early, authentic engagement with lab report writing and building student confidence through structured, feedback-driven planning.
Embedding AI tasks in digital quiz to enhance engagement and understanding of feedback	HC	Guide students to use GenAI constructively to interpret and apply feedback.	This approach uniquely leveraged GenAI to help students interpret feedback independently, enabling proactive correction of common

			lab report issues and reducing the need for one-on-one staff clarification.
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Table 1. Summary of Assessment activities described below and presented at workshop.

Activity Summaries

Introducing GenAI for Student Engagement and Literacy

Introducing Gemini to Bioscience Students

Liz Alvey, Sheffield University

What:

This intervention introduced bioscience students to large language models (LLMs), specifically Google Gemini, as tools for idea generation and critical thinking. A Gemini use case was integrated into an existing assessment and associated workshop. Students were asked to apply their biological knowledge to address the UN Sustainable Development Goals (SDGs), using LLMs to scaffold initial brainstorming and group discussion.

Why:

The activity was designed to embed digital literacy and responsible AI use within the bioscience curriculum, enabling students to become more critically reflective users of emerging technologies. By prompting students to critique AI-generated output, the session supported metacognitive development and collaborative learning (8). The task also supported faster project development and confidence in tackling complex global problems through structured ideation.

Where:

Second-year *Skills in Bioscience* module, delivered in the first of three one-hour workshops designed to support team-based projects addressing global challenges.

How:

- Transparent ground rules were established for using Gemini responsibly within the task (9).
- Students were instructed on how to access Gemini and given a ‘hello world’ example to build confidence.
- A customisable, structured prompt template was provided to scaffold early success (10).
- Students applied iterative prompting to a specific task, then critically reflected on the results as a team (11).
- **Prompt format:**
Act as a [CEO of a university biotech spin-out company that specialises in enzyme production]. List [10] ideas to tackle the UN’s Sustainable Development Goals. Produce different and varied plausible ideas. Link each example to a specific UN SDG.

Impact:

Students showed improved confidence in idea generation, accelerated their project development,

and gained a clearer understanding of both the strengths and limitations of LLMs. The interactive, device-flexible format created a dynamic and engaging workshop atmosphere, even in large cohorts. The task helped students engage meaningfully with global issues while building AI literacy.

Educator Quote:

"This was a fun, engaging way to introduce Gemini to all students. The atmosphere in the room was positive and very loud! There were no technical issues (even with very large cohorts), and students could fully engage with whichever device they brought with them. Have a go with your students when they next have an open-ended task—but allow some classroom time for initial exploration!"

Limitations:

Gemini may hallucinate references unless specifically prompted to link to reliable sources (Holmes et al., 2019). There is also a slight bias towards commercial over social solutions, which students were encouraged to critically evaluate.

Boosting student confidence through assessment literacy to reduce GenAI misuse

Juanvi Canet-Perez, University of Bradford

What: Level 6 Biomedical Science modules implemented strategies including assessment literacy workshops, staged coursework submissions, early formative drafts, and descriptive rubrics. These interventions aimed to deter unethical AI use by increasing student understanding of assessment criteria and fostering independent, authentic engagement.

Why: GenAI tools have become widely accessible to students over the past two years. As a result, universities are developing policies to preserve academic integrity in assessments. A critical review of 20 world-leading universities GenAI policies showed that their primary concern is around the originality of the students' work and the link between GenAI and academic misconduct (12). Universities have generally adopted a traffic light system for assessments where GenAI tools are used (green), can have a limited role (amber), or are not permitted at all (red). Whilst educators design new green and amber assessments, new strategies need to be implemented to *convince* students not to use GenAI tools in red assessments. The analysis of GenAI misuse (measured as reported cases of academic misconduct per assessment type) showed that reported cases were only associated with open book essays (11.3% and 17.5% reported cases in two modules). However, there was a third module with two assignments that fall in the same category (long written coursework submissions) and had no reported cases. Promoting transparency in assessment criteria is an inclusive pedagogical approach that supports students in becoming more independent and capable of self-assessment.

Where: Level 6 Biomedical Science students at Bradford undertake different types of assessments during their final year across five modules. These strategies were introduced in Biomedical Science Research Project module (2023/24, 80 students), and similar approaches were adopted across other Level 6 modules in 2024/25 (Medical Genetics and Research Topics).

How:

- Incorporated assessment literacy workshops where students used rubrics to mark anonymised past submissions.
- Collected anonymous student feedback (via MS Forms, % = Agree/Strongly Agree):
 - "I have a better understanding of assessment literacy and rubrics" – 91%

174 • “My performance will benefit from attending this session” – 93%
 175 • “I found this session useful” – 93%.

176 Broke coursework into smaller, manageable tasks (e.g., a 1000-word background and plan split into
 177 seven 100–200-word sections).

178 • Required submission of partial formative drafts to trigger academic concern meetings if
 179 GenAI misuse was suspected.
 180 • Developed **descriptive rubrics** including “poor academic practice” in the lowest mark band
 181 for all criteria, acting as a deterrent and offering markers flexibility to respond to suspected
 182 misuse.

183 **Students Quotes** (assessment literacy workshops): ‘Glad we got to see examples, this was very
 184 helpful and made me feel more confident and have an idea of what to write’, ‘Seeing the examples
 185 and talking about them really helped’, ‘The session was very informative’.

186 **Impact:** Boosting students’ confidence ahead of any assessment is a powerful tool to reduce GenAI
 187 misuse and to increase student performance. Student voice influenced the introduction of similar
 188 assessment literacy sessions in other modules.

189

190 **AI-generated MCQs and learning resources**

191 *Pedro Barra, Kingston University*

192 **What:** GenAI used to develop workshop materials and write assessment rubric

193 **Why:** The integration of GenAI into teaching materials and formative assessment design aimed to
 194 both save staff time and enhance student learning, particularly the development of critical thinking
 195 and data interpretation skills. These skills align with the module’s learning outcomes and support
 196 deeper engagement with complex nutrition research.

197 Using GenAI to generate multiple-choice questions (MCQs), rubrics, and learning resources supports
 198 constructive alignment, ensuring that learning activities and assessments are directly linked to
 199 intended outcomes. The use of AI to intentionally embed errors and challenge students to evaluate
 200 outputs promotes critical AI literacy and helps students move beyond passive consumption of
 201 content toward active analysis and evaluation, key higher-order skills in Bloom’s taxonomy.

202 This approach also supports assessment for learning principles by providing immediate, actionable
 203 feedback through platforms like Quizizz. Formative quizzes help reinforce retrieval and metacognitive
 204 reflection, especially when combined with student-driven interaction with LLMs. Encouraging
 205 students to critique and question AI-generated content builds feedback literacy and promotes
 206 a more autonomous and reflective learning approach.

207 **Where:** Year 2 BSc Nutrition, 30-credit applied nutrition module. LO to interpret and evaluate
 208 nutrition research. Supported by lecture and workshops in preparation for coursework screencast
 209 assignment.

210 **How:**

211 • Quizizz AI platform used to generate a formative assessment activity (in MCQ format) from
 212 existing educational resources uploaded to the platform.

- Students complete above task within the Quizizz AI platform (rather than on paper) before being presented with worksheets with applied exercises focusing on interpreting meta-analyses, which were generated using a Large Language Model (one that allows uploading images as part of a prompt). Students asked to check AI-generated summaries for accuracy and completeness.
- As extension exercises, we also asked GenAI to 1) include errors in its summaries and 2) create test MCQ questions. Students were also advised that they could use a Large Language Model (LLM) (Microsoft Copilot, licensed for student use) to test their understanding of images (forest plots) by asking the LLM to ask them questions and provide feedback on their answers.
- GenAI used to create an assessment rubric for the coursework screencast summative assessment, which tested interpretation and evaluation of nutrition research, for which the activities above helped prepare students for.

Impact: Completion of quizzes online facilitates generation of real-time results, thus allowing for tailored and actionable feedback to be provided to students, before students move onto the next task. Writing learning and teaching resources, as well as developing assessment rubrics, are time-demanding activities, where the use of GenAI can offer time-savings and help extend the scope of teaching materials. The module results for this assessment task improved compared to previous years, although the assessment task was originally run under examination conditions and is now coursework. However, the task is exactly the same as in the exam (i.e., the mode of assessment changed but the question format and associated task are the same).

Student Quote

“Using the GenAI-based resources really helped me understand how to describe and interpret data more effectively. The instant feedback from the quizzes and the structured worksheets based on real research papers made complex statistical concepts much clearer and more engaging. It also gave me confidence in analysing figures and tables independently, which supported my learning massively.”

Using GenAI to Enhance Teaching Resources and Learning Design

Rapid prototyping of learning resources

Chris Hassall, University of Leeds

What:

In a postgraduate module on *Equitable Sustainability*, students engaged in a dynamic, interdisciplinary learning process that included prompt lectures from external experts followed by seminar discussions. When students expressed interest in a previously unaddressed topic, strategic litigation in conservation biology, the educator used ChatGPT to rapidly generate a mini-lecture. Within just over an hour, a 10-slide deck was developed that blended AI-generated content with the educator’s disciplinary framing. This provided timely, relevant input for the next day’s seminar.

Why:

This intervention was motivated by a desire to respond meaningfully to emergent student interests and to support learner agency in shaping the curriculum. The educator’s goal was to model

responsive pedagogy and provide a scaffolded resource that enriched student-led inquiry without requiring subject-matter expertise in the new topic. GenAI enabled the educator to bridge that gap quickly, while maintaining a critical eye on factual accuracy (Zawacki-Richter et al., 2019). The task also supports curriculum as dialogue, reinforcing active and participatory learning (Nicol & Macfarlane-Dick, 2006).

Where:

The module (*BLGY5231M: Equitable Sustainability*) is attended by students from a wide range of disciplinary backgrounds. The specific session followed a one-hour guest lecture by a lawyer, and the educator-facilitated follow-up seminar the next day allowed students to delve deeper into the new topic.

How:

- An initial broad prompt was entered into ChatGPT: “*Make a 10-slide deck about strategic litigation in conservation biology.*”
- A follow-up prompt, “*Can you make a PowerPoint file with those slides?*”, generated a formatted presentation. A visual theme was added to enhance design.
- Further prompts were used to populate the slides with relevant examples. For instance, “*Can you describe a landmark case in climate change litigation?*” produced a summary of *Urgenda Foundation v. The State of the Netherlands*, while “*Do you have an equivalent in biodiversity conservation?*” returned details of *Northern Spotted Owl v. Hodel*. Each was fact-checked and supplemented with additional academic sources before presentation.
- **Link:** <https://chatgpt.com/share/100bea94-58d6-41ab-af81-f8c2c1eb619f>

Impact:

This approach reduced the cognitive and logistical load of producing new learning materials, enabling faster response to students' interests without compromising quality (10). The AI-supported resource enhanced student engagement by providing structured, contextualised input that supported deeper discussion. Students came to the seminar better informed and more confident, contributing to a rich, critical dialogue around the topic. The process modelled how GenAI can support student-led and adaptive learning in fast-paced, interdisciplinary contexts.

Educator Quote:

"Using ChatGPT helped me meet students where their interests were. It saved time, but more importantly, it allowed for student-centred flexibility in a fast-paced, interdisciplinary course."

Generation of ideas for active learning

Chris Hassall, University of Leeds

What: GenAI (ChatGPT) was used to identify active novel learning activities associated with a lecture, beginning with several diverse initial ideas and then a detailed sketch of a selected activity including images. In this particular case, the students were attending a lecture on algal blooms as a “grand challenge” in biology and so the educator was interested in a more detailed discussion of potential solutions rather than a didactic lecture.

Why: The aim was to enhance student experience and engagement in an otherwise lecture-based course by integrating elements of active and problem-based learning. By framing algal blooms as a

real-world, interdisciplinary challenge, the learning activity encouraged critical and creative thinking. This approach is grounded in constructivist learning theory (8) which emphasizes that students construct knowledge actively through interaction and reflection. GenAI acted as a cognitive tool, supporting the educator in the design process and enabling rapid ideation and iteration of authentic, inquiry-driven tasks.

In doing so, the educator also embraced the role of a learning designer and curator, using ChatGPT not to replace pedagogical expertise but to enhance it, prompting metacognitive reflection on how to foster meaningful engagement. This aligns with evidence supporting active learning strategies, which have been shown to improve student performance in STEM disciplines (13). The use of GenAI in this context thus serves not only to facilitate more engaging lectures but also to model the use of emerging technologies in innovative, educationally grounded ways.

Where: BLGY1208 Living Planet is a compulsory module that takes students from the origins of life to sustainability research with many academics teaching one or a few lectures. This academic had only one 1-hour lecture slot with the students to cover the causes and consequences of algal blooms and so chose to use a flipped classroom (4 x 10-minute videos followed by the 1 hour of contact time).

How:

- An initial prompt was used to generate a list of ideas for active learning: *I have a 1st year undergraduate lecture to give to biology students and wish to use active learning. The topic is "harmful algal blooms" Can you suggest some interactive activities?*. That list included a range of eight techniques from debates to field trips.
- Having identified a role playing example as one to pursue further, a second prompt was used to ask for an outline of such an activity: *"I like the role playing simulation, could you sketch that out in more detail, with a guide for the students?"* ChatGPT responded by providing an objective, a scenario (describing the fictional location of "Aqua Bay"), a list of roles, and instructions for the students. While the described activity was too long for the 1-hour session, a reduced form was useful for 40 minutes of discussion.
- To enrich the activity, a prompt was used to request an image of Aqua Bay to show the students what such a location might look like: *"could you generate an image of "Aqua Bay" showing the algal bloom, please?"*.

Impact: Co-creation of active learning materials with GenAI sped up activity design and allowed the lecture to be more creative. Despite misgivings about how the students would interact with the activity, there was considerable engagement with the role playing that showed learning during the discussions.

Quote: "[The active learning approach] manages to get a lecture theatre of usually silent students to discuss and explore ideas surrounding conservation. [The activity] makes it a safe environment to put yourself out there..."

Prompt engineering for lecture transcripts

Chris Hassall, University of Leeds

What: GenAI (Chat GPT) was used to develop lecture summaries from the transcripts automatically generated during recorded lectures. These lecture summaries included a summary of the points covered, suggestions for further reading, and questions for consideration by students to develop their understanding.

Why: Lecture recordings and their associated transcripts are an underused educational resource. By applying GenAI to extract key concepts, signpost content, and generate questions, educators can reduce the extraneous cognitive load on students (14), freeing up time for deeper engagement with the subject through reading and reflection. These summaries act as a scaffolding tool guiding learners through complex material and supporting self-regulated learning ((15). Additionally, this approach aligns with the principles of Universal Design for Learning (16), enhancing accessibility and accommodating diverse learning needs. The inclusion of guiding questions fosters metacognitive awareness (14), helping students think critically about their learning process. Overall, using GenAI in this way exemplifies how digital resources and AI-driven tools can personalise and enhance education by turning passive content into an active learning resource.

Where: Lectures are recorded using the University of Leeds lecture capture tool with automatic captions. A single 1-hour lecture was chosen as a test case from a 1st year module.

How:

- The transcript of the lecture was uploaded to ChatGPT in order to produce a cleaner summary of the content that was covered: "I have attached the transcript of a lecture that I gave. Please could you tidy the language where the transcription was wrong and export it as a txt file."
- Since this lecture transcript was too long (ca. 5,000 words), a summary was also created: "*summarise the lecture content to pull out the main themes covered*". The result was ca. 420 words as bullet points.
- ChatGPT 4.0 hallucinated extensively when asked for outside reading to supplement the lectures. Instead, ScholarGPT was used to search scientific databases using the following prompt: "*Can you suggest three papers written in the last three years that are relevant to this lecture content, all in Q1 journals, and provide their references in Harvard format.*"
- Finally, ChatGPT was asked to provide some topics that the students could consider further to develop their own understanding of the subject: "*Can you give three prompts for further consideration for the students based on this attached lecture material?*"

Impact: Lecture summaries will be deployed ahead of the lecture delivery in the 2025/26 academic year.

Quote: This approach was prompted by comments on lecture volume, clarity, and complexity, such as this paraphrased quote from my Faculty National Student Survey comments: "*Lecturers sometimes mention concepts that seem significant but don't explain them in detail. They may also present too many slides or ideas in a single lecture, making it difficult to follow along and fully grasp the material.*"

Using Integrated Knowledge Prompting to develop written content using large language models.

Matthew Hardy, University of Bradford

What: Liu and colleagues (17) developed generative knowledge prompting for getting common-sense output from large language models (LLMs). This two-step process first generates multiple knowledge statements from few-shot prompts in one language model, before using a second language model to integrate these statements. The proposed methodology adapts this process by replacing the first step with human knowledge input. The LLM then integrates this knowledge to produce output. In the 'how' described below, the human knowledge input consists of language structure; followed by

critical appraisal and factual information the author has gained from journal articles. The LLM then integrates this to provide written content.

Why: There are two reasons for using integrative knowledge prompting. Firstly, this approach takes advantage of the strengths provided by LLMs, whilst negating some of their flaws. Namely, we can rely on these models to produce grammatically accurate text, but the knowledge they produce is fallible because they are prone to 'hallucinate'. Secondly, there are barriers to writing that LLMs can help to overcome. These include dyslexia, dysgraphia and writing English as a second language. For example, non-native English speakers may take 30-50% more time to write an academic paper and are 2.5 times more likely to have a paper rejected specifically for writing issues (18).

Where: Academic Skills & Professional Development Workshops, University of Bradford.

How:

- Instruct an LLM (e.g. ChatGPT 3.5) to use a specific structure (e.g. paragraph structure: the number of sentences to be included in a Claim, Justification, Support and Conclusion) to generate written content from 'knowledge' you will provide.
- Supply the LLM with 'knowledge' derived from relevant journal articles. This may include factual content (e.g. 'Knowledge: GPIIb/IIIa antagonists speed up the lysis of clots') and critical appraisal (e.g. 'Knowledge: Human data is more reliable for treatment of stroke than rat data'). You should include journal citations so they can be used for referencing.
- Refine output using prompt chaining (e.g. 'Rewrite the last paragraph but do not make any statements that are not confirmed with a citation. Use the structure outlined in the first prompt. Format: academic')

Impact: This workflow encourages students to understand the knowledge that they are using, whilst removing some barriers to producing written work.

Quote: 'Unless learning how to write is an objective, this is a valid use of a tool for science communication.' - Staff member

Designing Authentic and AI-Resilient Assessment and Feedback

Creation and Design of Graphical Abstracts as an Assessment

Andre Pires-da Silva, University of Warwick

What: This assessment involves asking students to produce a graphical abstract from a paper and justify the choice of elements. A graphical abstract is defined as a pictorial summary of the main findings of a research paper, often used by journals to highlight key points concisely. For example, in a physiology context, students could be assigned a paper exploring the relationship between diet restriction and rate of ageing. Their graphical abstract may illustrate how insulin signalling changes with diet restriction and how insulin resistance results in age-related diseases. Such a task requires students to distil a complex interplay of molecular, physiological, and systemic factors into a concise visual summary.

Why: The assessment is designed to measure deep conceptual understanding and the ability to synthesise and communicate complex information, skills that align with the higher-order levels of Bloom's Taxonomy, including analysis, evaluation, and creation. In the diet restriction and ageing

example, students must integrate knowledge of metabolic regulation with evidence on its implications for organismal lifespan, identify the essential components of feedback loops, and visually convey how perturbations in these systems influence ageing trajectories. This goes beyond recall or comprehension, requiring students to apply content knowledge in a creative format that demands both scientific literacy and visual communication skills.

This form of assessment also addresses the limitations of Generative AI, which currently struggles with creativity and contextual accuracy. While GenAI tools are effective at supporting lower-order cognitive skills such as summarisation or basic recall (remembering and understanding), they perform poorly in tasks requiring original, biologically accurate visual synthesis, often producing misleading or incorrect imagery and text. As such, the task reinforces the value of human creativity and disciplinary expertise, a key pedagogical principle in the age of AI-enhanced education (19).

Additionally, the graphical abstract assessment supports dual coding theory (20), which posits that learning is enhanced when information is processed both visually and verbally. By requiring students to translate scientific knowledge into visual form, the task fosters deeper cognitive integration and improved retention of information.

Where: This assessment was implemented in the third-year module Developmental Genetics (BS373) at the University of Warwick.

How:

Students should create a clear and informative graphical abstract by following these guidelines:

- Use primarily visual elements to communicate the core findings of the paper.
- Ensure the abstract clearly conveys the main conclusions, rather than re-presenting data.
- Include a brief written justification explaining the rationale behind the choice of visual components.
- Design the abstract to have a logical visual flow—either top-to-bottom or left-to-right.
- Provide sufficient context for the results, such as indicating tissue type, organism, or experimental setting.
- Avoid reusing figures from the original paper; instead, focus on conceptual representation of new findings.
- Exclude excessive detail, speculative interpretations, or background from prior literature.
- Use simple labels and minimal text, avoiding visual clutter or unnecessary embellishment.

Impact: This type of assessment is easy to mark and is sustainable, as it is easy to make new questions by selecting different research papers. Grading these assignments is efficient, taking only a few minutes per submission. More importantly, it ensures that students not only have full understanding of the paper in question, but also develop creativity and design skills to appropriately convey the main concepts in a simple pictorial form.

Educator Quote: “This assignment requires cognitive skills beyond the capabilities of current AI systems. Looking ahead, other visual assessment formats like short video productions could further challenge AI in this domain. While graphical abstract creation services exist, they are currently time-consuming and expensive.”

Comparing two approaches to SAQs: adapt or adopt to AI

Denise Hough, University of Glasgow

What: Implemented and compared different approaches across two bioscience courses that either adopted or adapted to generative AI, given the prevalence of online Short Answer Questions (SAQs) in their summative assessments.

Why: SAQs are widely used in early-stage bioscience education because they promote retrieval practice and help assess students' foundational understanding. However, their transition to online open-book formats, accelerated during the COVID-19 pandemic (21), has introduced challenges around assessment validity and the authenticity of student responses (22). The rapid advancement of GenAI has further amplified these concerns, as such tools can generate plausible but unoriginal answers, undermining the ability of SAQs to evaluate individual learning effectively.

In response, educators must rethink assessment design to preserve construct validity, promote academic integrity, and support assessment for learning rather than merely assessment of learning (23). This includes combining SAQs with robust formative tasks, designing questions that require higher-order thinking, and engaging students in transparent discussions about the ethical use of AI in their studies.

Where: At the University of Glasgow, GenAI-supported teaching activities were implemented in two undergraduate courses: the first-year *Veterinary Biosciences* course VETSCI1006: Digestive Physiology and Metabolism, with approximately 30 students, and the second-year *Life Sciences* course BIOL2041: Animal Biology, Evolution and Ecology, which enrolled around 200 students.

How:

- Interactive Session: Both courses included a session on the [responsible use of generative AI in higher education](#).
- Adapt to AI (VETSCI1006): Transitioned from online open book SAQs (1 formative, 1 summative) to digital closed book (invigilated, summative class test and exam).
- Adopt AI (BIOL2041): Maintained online open book SAQs for the class test, supplemented by an AI-powered revision chatbot ([Mizou](#)) providing instant feedback. The SAQ exam section was replaced by extended multiple choice questions (eMCQs), requiring critical thinking and knowledge application. Performance in eMCQs was comparable to pre-pandemic closed book SAQs, and generative AI did not outperform students.
- Assessment and Feedback: Multiple markers graded SAQs and provided written feedback. Approximately 22% of answers were flagged for suspected non-authenticity, which decreased to 4.5% after review by the course coordinator, who then offered additional feedback and one-to-one meetings (50% uptake).

Impact: In *VETSCI1006*, students performed well on open-book assessments but struggled with closed-book exams, despite a gradual transition between formats. This highlights the need for more robust preparation strategies to support knowledge retention and exam readiness. In *BIOL2041*, the short-answer question (SAQ) chatbot received highly positive student feedback, suggesting it effectively supported learning. However, concerns remain about the authenticity of student responses in online open-book assessments, even with prior education on generative AI and its appropriate use. These findings suggest that while electronic multiple-choice questions (eMCQs) may be suitable for online open-book formats, SAQs are more appropriately assessed in closed-book

settings, reinforced by frequent formative tasks. Maintaining open dialogue with students about the use and limitations of GenAI remains essential.

Educator Quote:

"We must remember that no single item of assessment should exist in isolation; it is part of a continuum of complementary activities, including both formative and summative tasks. I think that generative AI can be extremely useful in formative tasks to build foundational knowledge in biosciences, but I also think that many students will be less motivated to internalise that knowledge if they have access to generative AI during summative tasks, which will ultimately undermine those students' learning in the long run."

Using digital quizzes as formative assessments to engage and prepare students for lab reports

Hannah Campbell, University of Leeds

What: A digital interactive quiz was developed as a formative assessment tool to support students in preparing lab reports. The quiz included embedded snippets of anonymised work from previous students, which served as examples and interactive activities. This allowed learners to evaluate, critique, and compare writing samples to build their understanding of lab report expectations.

Why: This approach encourages active learning by allowing students to practise and reflect in a low-stakes, engaging environment. The quiz provides immediate feedback, a key factor in promoting learning and supporting formative assessment practices. By engaging with real student examples of varying quality, learners develop the ability to distinguish between strong and weak academic writing, fostering assessment literacy (24) and self-regulation (11).

Moreover, using this tool as a reusable resource across the academic programme aligns with principles of sustainable assessment, where formative tools contribute to long-term learning beyond a single task (24). It also supports confidence building and constructive alignment (9) helping students see how assessment criteria apply in practice.

Where: University of Leeds

How:

- Each quiz was designed around clearly defined learning outcomes, aligned with the skills needed for effective lab report writing.
- Anonymised excerpts of previous student work were integrated to serve as examples for evaluation and discussion.
- The quiz was hosted on the LT Kuracloud platform, which allows for interactive, feedback-rich experiences; similar functionality could be replicated on other digital platforms.
- Generative AI was used to assist in drafting questions and feedback, supporting efficient content creation while retaining pedagogical oversight.

Impact: The quiz reduced staff marking workload by automating formative feedback and encouraging self-directed learning. It served as a deterrent against inappropriate GenAI use, as students built genuine confidence through trial-and-error learning in a low-stakes environment. As a reusable resource, it continues to support skill development across multiple stages of the programme.

A Lab Report Plan as a formative assessment task for a Summative Lab Report

Hannah Campbell, University of Leeds

What: Students completed a *Lab Report Plan* as a formative assessment, submitted ahead of their full lab report. This plan was reviewed by markers (typically Personal Tutors), who provided targeted feedback. The plan was also linked to a digital interactive quiz, offering additional guidance and feedback aligned with key learning outcomes.

Why: This approach encourages scaffolded learning, helping students break down a complex task into manageable components, such as selecting appropriate statistical analyses, planning figures and tables, identifying relevant references, and drafting key sections. It also fosters self-regulated learning by prompting students to think in advance about what they will do and how they will approach it (15). Lab reports are a cornerstone of physiology education, as they develop essential skills in experimental design, data interpretation, and the articulation of complex biological processes such as cardiovascular regulation or neurophysiological responses

Providing feedback on the plan, rather than the final product, supports feedforward pedagogy (24), which focuses on improving future performance rather than merely justifying a grade. When paired with linked resources and guidance, this method promotes formative assessment for learning (23), enabling students to act on feedback, develop writing confidence, and reduce cognitive load when drafting their full report (14).

Where: University of Leeds

How:

- The Lab Report Plan was developed in conjunction with the digital quiz to provide a cohesive, scaffolded learning experience.
- Each section of the plan aligns with the corresponding section of a full lab report, guiding students step-by-step through the process.
- Embedded key questions prompt students to consider both *what* they will include and *why*, encouraging critical thinking and intentional planning.

Impact: The approach reduced staff marking workload by shifting feedback earlier in the process and supporting self-directed, iterative improvement. It acted as a deterrent to inappropriate GenAI use by fostering genuine confidence through structured trial and error. The resource also offered clear signposting to further guidance, increasing student engagement with formative tasks and improving preparation for summative assessment.

Embedding AI tasks in digital quiz to enhance engagement and understanding of feedback

Hannah Campbell, University of Leeds

What: Interactive quizzes were enhanced by embedding targeted questions and guidance that prompted students to use generative AI to help interpret and expand on feedback received from previous lab reports. The aim was to encourage students to engage more deeply with feedback by asking GenAI to explain unclear comments, suggest ways to improve, or model how revised sections might look.

Why: Students often struggle to interpret written feedback and translate it into concrete improvements. Embedding AI-guided reflection into a structured quiz format supports feedback literacy (24), helping students better understand the purpose and application of feedback. By actively engaging with their own feedback using GenAI as a conversational partner, students develop self-regulation and metacognitive awareness (11).

The use of interactive quizzes as a scaffolded environment for this reflection aligns with formative assessment principles (23) offering low-stakes, iterative opportunities to practice improving academic work. Furthermore, prompting students to use GenAI in a focused, transparent way supports ethical AI literacy, equipping them to use these tools constructively rather than inappropriately (19).

Where: University of Leeds

How:

- Embedded examples of anonymised past student work, along with associated tutor feedback, were used within a digital quiz.
- Students were prompted to copy feedback comments and use a specific GenAI prompt, such as *“Can you elaborate on this feedback?”*, to encourage deeper interpretation.
- GenAI was used to help students break down feedback into key themes, highlighting what was done well and identifying areas for improvement in a clear, student-friendly manner.

Impact: This approach reduced staff workload, as individual feedback clarification sessions were not feasible for large cohorts. It enhanced students’ understanding of feedback and demonstrated a productive, ethical use of GenAI. The method also supported proactive learning, addressing common lab report issues (e.g., incorrect tone, protocol-style methods, repetition in discussions) before submission.

Conclusion

The case studies presented in this paper highlight the diverse ways generative AI is being integrated into teaching, learning, and assessment in higher education. Across a range of contexts, educators are using these tools to create learning materials more efficiently, support student engagement, and experiment with new assessment formats that prioritise process, authenticity, and student development.

A recurring theme is the use of generative AI to promote active and student-centred learning. In many examples, generative AI supported formative activities that encouraged reflection, iteration, and the development of higher order thinking skills. Rather than focusing solely on end products, these approaches valued the steps students took to reach their outcomes, aligning with broader shifts toward process-oriented assessment.

Generative AI was also shown to enhance feedback and assessment literacy, enabling students to better understand expectations and apply feedback in meaningful ways. By integrating generative AI into formative tasks, digital quizzes, and assessment preparation, educators supported students’ confidence and autonomy while reducing staff workload.

However, the limitations of generative AI remain clear. Its reliance on existing data and difficulty with complex reasoning or creative synthesis present challenges for certain assessment types and its

difficulty with tasks that require high-level creative and critical thinking (19);(6). Several case studies demonstrated the need for assessment formats that are resilient to AI misuse, such as visual tasks, closed conditions, or assessments that require contextualised and original responses.

The examples reinforce the importance of transparency, ethical guidance, and structured support when using generative AI in education. Embedding digital and assessment literacies into curricula not only reduces inappropriate use but also helps students become more responsible and critical users of AI tools.

Overall, these case studies demonstrate that with thoughtful implementation, generative AI can complement established pedagogies and support more inclusive, engaging, and reflective learning experiences. Continued experimentation, collaboration, and evaluation will be essential to ensure that the use of generative AI in education enhances rather than undermines the values of academic integrity, creativity, and deep learning.

While there are hurdles to overcome, the potential of GenAI as an enabler in higher education is undeniable. With careful implementation and ongoing evaluation, GenAI can significantly enhance the student learning experience, making education more accessible, engaging, and effective. As we continue to harness this potential, the future of higher education looks poised to be more innovative, inclusive, and insightful.

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